MILITARY SPECIFICATION

INSTRUMENTS, SIMULATED, AIRCRAFT TRAINING DEVICES; DIRECTIONAL GYRO

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The technical requirements for procuring the instrument described herein shall consist of this document and the issue in effect of Specification MIL-I-82356.

1. SCOPE

1.1 This specification covers the detail requirements for simulated aircraft directional gyro indicators for use in aircraft training devices.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-5126	Indicator, Directional, Gyroscopic, Magnetic Compass, Type V-7A
MIL-I-5741	Indicator, Directional, Gyroscopic, Type C-5C
MIL-I-7803	Indicators; Directional Gyro, Electric
MIL-S-20708	Synchros, General Specification For
MIL-I-82356	Instruments, Simulated, For Aircraft Training Devices, General Specification For

STANDARDS

Military

MS 25495	Indicator - Directional Gyro, Electric

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Procuring Contracting Officer.)

3. REQUIREMENTS

3.1 Operational instruments.- The requirements for the operational instruments being used as simulated instruments shall be as specified in the operational instrument specifications referenced in Table II.

3.2 Simulated instruments.- The requirements of MIL-I-82356 and this specification shall apply.

3.2.1 Design basis.- Each simulated aircraft and operational aircraft directional gyro indicator shall conform to the applicable operational instrument specified in Table I of this specification. In each case for Table I, the activating component shall be coupled to the pointer with a 1:1 ratio.

3.2.2 Case.- With the exception of length which shall not exceed the limits specified in the footnotes, cases shall be in accordance with Table I of this specification, as applicable.

3.2.3 Wiring.- Wiring shall be in accordance with schematics referenced in Table I as applicable.

3.2.4 Setting knobs.- Hand operated knobs identical in appearance and function to the hand operated knobs on the operational instrument for setting the dial and pointer shall be provided for each simulated instrument, as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-I-82356 and as specified in 4.2 through 4.3.

4.2 Tests.- The following tests shall be performed on each instrument referenced in Table I in addition to tests specified in the test requirements of MIL-I-82356.

(a) Backlash of gears

(b) Forward and reverse operation

(c) Running smoothness of gears

(d) No jumping or hesitation of pointers.

4.2.1 Electrical tests.- Electrical tests for instruments referenced in Tables I and II shall consist of checks on electrical zero and running performance. Test equipment shall consist of a transmitter synchro of

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a size equal to or larger than that employed in the instrument actuator to be tested. The test transmitter synchro shall be rotated by a precision dial where zero degrees has been made to correspond to the reference line at the transmitter's true electrical zero in accordance with MIL-S-20708.

4.2.1.1 Electrical zero.- Tests for electrical zero shall be conducted with the test transmitter synchro transmitter phase coordinated with the activated receiver synchro in accordance with MIL-S-20708. When the test transmitter synchro dial is at $0^{o} \pm 0.5$ degrees, each activated receiver synchro pointer shall read its electrical zero position $\pm$ 1.0 degree.

4.2.1.2 Running tests.- Each receiver synchro, connected in proper phase compatibility as outlined in MIL-S-20708, shall be subjected to the following running tests using the test set-up specified in 4.2.1.

(a) Slow run - Each transmitter synchro shall be rotated clockwise and then counterclockwise at approximately 1 rpm for a minimum of 1.5 turns. The receiver synchro indicator pointer shall track smoothly without any discontinuities

(b) Fast run - The test of 4.2.1.2(a) shall be repeated a minimum of 3 revolutions in either direction at a rate of approximately 1.5 revolutions per second. The receiver synchro display pointers shall follow without any discontinuities and not lag the transmitter synchro display pointer by more than 10 degrees.

4.3 Operational instrument tests.- The test procedures for instruments referenced in Table II shall be in accordance with the test procedures of the applicable operational specification listed in Table II. These tests shall be performed under the test requirements of MIL-I-82356.

5. PREPARATION FOR DELIVERY

5.1 The preparation for delivery requirements of MIL-I-82356 shall apply.

6. NOTES

6.1 The notes of MIL-I-82356 shall apply.

CUSTODIANS:
ARMY - AV
NAVY - TD
AIR FORCE - 70

USER ACTIVITIES:
ARMY - AV
NAVY - TD
AIR FORCE - 11, 70

PREPARING ACTIVITY:
NAVY - TD
Project Number 6930-0068

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TABLE I - SINGLE SYNCHRO DATA

DESIG-NATION	OPERATIONAL INSTRUMENT BEING SIMULATED	NOMINAL SIZE	*CASE -TYPE	SCHEMATIC WIRING MIL-I-82356 FIG. NO.
-1 S	MIL-I-5741**	4 1/2 inch	MIL-I-5741	4
-2 S	MIL-I-7803		MS 25495	4

* Maximum case length of simulated instruments may exceed the case length specified in the specifications noted above. However, the maximum case length should not exceed 6 inches without prior approval from the procuring activity.

** The precession rate of the compass dial if specified shall be provided by a 26V DC 1/4 RPH timing motor.

An ON-OFF flag and caged flag are to be provided which will appear identical to the operational instrument and shall be activated by 26V DC with a maximum current consumption of 10 MA.

TABLE II - OPERATIONAL INSTRUMENTS

DESIG-NATION	OPERATIONAL INSTRUMENT BEING USED IN SIMULATED EQUIPMENT	NOMINAL SIZE	CASE TYPE	INTEGRALLY LIGHTED
-1 A	MIL-I-5126	5 inch	MIL-I-5126	NO

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